

EXHIBIT 5

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

SCHEDULE 14A
(Rule 14a-101)

SCHEDULE 14A INFORMATION

Proxy Statement Pursuant to Section 14(a) of the Securities Exchange Act of 1934
(Amendment No.)

Filed by the Registrant ☒
Filed by a Party other than the Registrant ☐
Check the appropriate box:

- ☐ Preliminary Proxy Statement
- ☐ **Confidential, for Use of the Commission Only (as permitted by Rule 14a-6(e)(2))**
- ☒ Definitive Proxy Statement
- ☐ Definitive Additional Materials
- ☐ Soliciting Material Pursuant to §240.14a-12

Theriva Biologics, Inc.
(Exact Name of Registrant as Specified in its Charter)

(Name of Person(s) Filing Proxy Statement, if other than the Registrant)

Payment of Filing Fee (Check the appropriate box):

☒ No fee required.

☐ Fee paid previously with preliminary materials.

☐ Fee computed on table in exhibit required by Item 25(b) per Exchange Act Rules 14a6(i)(1) and 0-11.



**NOTICE OF SPECIAL MEETING OF STOCKHOLDERS
TO BE HELD ON FEBRUARY 11, 2026**

Notice is hereby given that Theriva Biologics, Inc. ("**we**," "**us**," or the "**Company**") will host a Special Meeting of Stockholders on February 11, 2026, at 3:30 p.m. (local time) (the "**Special Meeting**"). The Special Meeting will be held in person at the Company's principal place of business for research and clinical development, located at Carrer Torrent de Can Ninou, naus 5-6, 08150 - Parets del Vallès (Barcelona) Spain. The Special Meeting is being held for the following purposes:

- (1) To approve, pursuant to Section 713(a) of the NYSE American Company Guide and the terms of an inducement agreement, the issuance of up to an aggregate of 16,184,560 shares of our common stock upon the exercise of our common stock purchase warrants issued to institutional investors in our private placement offering, which closed on October 17, 2025 (the "**Warrant Exercise Proposal**"); and
- (2) To approve a proposal to adjourn the Special Meeting to a later date, if necessary or appropriate, to permit further solicitation and vote of proxies in the event that there are insufficient votes for, or otherwise in connection with, the approval of the Warrant Exercise Proposal (the "**Adjournment Proposal**")

Stockholders of record at the close of business on December 31, 2025, are entitled to notice of and to vote at the Special Meeting and any postponements or adjournments thereof. A list of stockholders of record at the close of business on December 31, 2025, will be available for inspection by any stockholder for a period of ten days prior to the Special Meeting at our principal office at 9605 Medical Center Drive, Suite 270, Rockville, Maryland 20850.

Your vote is very important. Whether or not you plan to attend the Special Meeting, we encourage you to read the accompanying Proxy Statement and to submit your proxy or voting instructions as soon as possible. In order to ensure the representation of a quorum at the Special Meeting, stockholders who do not expect to attend the Special Meeting are urged to vote as soon as possible. For information on how to vote your shares, please refer to the section of the Proxy Statement entitled "Questions and Answers About the Proxy Materials and the Special Meeting" and to the instructions provided in your proxy card or by your broker, bank, or other nominee.

By Order of the Board of Directors:

/s/ Steven A. Shallcross

Steven A. Shallcross

Chief Executive Officer,

Chief Financial Officer and Director

Dated: January 5, 2026

**Important Notice Regarding the Availability of Proxy Materials
for the Stockholder Meeting to Be Held on February 11, 2026**

The proxy statement is available at www.proxyvote.com.



Theriva Biologics, Inc.
 9605 Medical Center Drive, Suite 270
 Rockville, Maryland 20850

**PROXY STATEMENT
 FOR THE SPECIAL MEETING OF STOCKHOLDERS TO BE HELD ON FEBRUARY 11, 2026**

GENERAL INFORMATION

The Board of Directors (the “**Board**”) of Theriva Biologics, Inc., a Delaware corporation (the “**Company**”), is soliciting proxies to be used at a Special Meeting of Stockholders to be held on February 11, 2026, at 3:30 p.m. (local time) (the “**Special Meeting**”) at the Company’s principal place of business for research and clinical development, located Carrer Torrent de Can Ninou, naus 5-6, 08150 - Parets del Vallès (Barcelona) Spain. This proxy statement (this “**Proxy Statement**”) and the accompanying proxy card are posted on the internet at www.proxyvote.com, and will be distributed and made available to our stockholders on or about January 5, 2026. If you previously requested electronic delivery of the proxy materials, you will be sent the proxy statement and the accompanying proxy card on or about January 5, 2026.

Voting Matters and the Board’s Recommendation

Agenda Item	Board Vote Recommendation	Page Reference
Approval of the issuance of up to an aggregate of 16,184,560 shares of our common stock upon the exercise of our common stock purchase warrants issued to institutional investors in connection with our private placement offering, which closed on October 17, 2025 (the “Warrant Exercise Proposal”)	FOR	7
Approval of the adjournment of the Special Meeting to a later date, if necessary or appropriate, to permit further solicitation and vote of proxies in the event that there are insufficient votes for, or otherwise in connection with, the approval of the Warrant Exercise Proposal	FOR	10

Even if you plan to attend the Special Meeting, please vote in advance so that your vote will be counted if you later decide not to attend the Special Meeting.

**QUESTIONS AND ANSWERS ABOUT
THE PROXY MATERIALS AND THE SPECIAL MEETING**

Who is soliciting my vote?

The Board of Directors of Theriva Biologics, Inc. is soliciting your proxy to be used at the Special Meeting.

When and where will the Special Meeting be held?

The Special Meeting will be held on February 11, 2026, at 3:30 p.m. (local time), at the Company's principal place of business for research and clinical development, located at Carrer Torrent de Can Ninou, naus 5-6, 08150 - Parets del Vallès (Barcelona) Spain.

What do I need to do if I would like to attend the Special Meeting?

If you wish to attend the Special Meeting in person, you must present a valid form of photo identification, such as a driver's license. If you are a beneficial owner of common stock that is held of record by a bank, broker, or other nominee, you will also need proof of ownership to be admitted. In this regard, a recent brokerage statement or a letter from your bank or broker are examples of proof of ownership. The Company reserves the right to prohibit cameras, recording equipment, or electronic devices in the Special Meeting.

What am I voting on at the Special Meeting?

The purpose of the Special Meeting is to vote on the following items described in this Proxy Statement:

Proposal No. 1: To consider and approve the issuance of up to an aggregate of 16,184,560 shares of our common stock upon the exercise of our common stock purchase warrants issued to institutional investors in our private placement offering, which closed on October 17, 2025 (the "**Warrant Exercise Proposal**").

Proposal No. 2: To consider and approve the adjournment of the Special Meeting to a later date, if necessary or appropriate, to permit further solicitation and vote of proxies in the event that there are insufficient votes for, or otherwise in connection with, the approval of the Warrant Exercise Proposal (the "**Adjournment Proposal**").

Who is entitled to vote?

Stockholders as of the close of business on the record date of December 31, 2025 (the "**Record Date**"), are entitled to vote at the Special Meeting or any postponement or adjournment thereof. As of the Record Date, there were 35,688,349 shares of the Company's common stock, par value \$0.001 per share (the "**Common Stock**"), outstanding.

How many votes per share of Common Stock held are stockholders entitled to?

Stockholders have one vote per share on all matters presented at the Special Meeting.

What is the difference between holding shares of Common Stock as a "stockholder of record" and holding shares in "street name"?

Shares held as a "stockholder of record" (also called a "registered holder") are shares held directly in your name. Shares held in "street name" are shares held for you in an account with a broker, bank, or other nominee.

How do I vote my shares?

If you are a *registered holder*, you may vote:

- **By internet.** Via the Internet at www.proxyvote.com;
- **By telephone.** If you are located within the United States and Canada, call 1-800-690-6903 (toll-free) from a touch-tone telephone;
- **By mail.** By returning a properly executed proxy card in the postage-paid envelope provided or return it to Vote Processing, c/o Broadridge, 51 Mercedes Way, Edgewood, NY 11717; or
- **In person.** You may vote in person at the Special Meeting.

To vote online or via telephone, you will need your unique control number. You can find the control number on your proxy card. Be sure to have your proxy card in hand and follow the instructions. Internet and telephone voting facilities will close at 11:59 p.m. Eastern Time on February 10, 2026, for the voting of shares held by stockholders of record as of the Record Date. Proxy cards with respect to shares held of record must be received no later than February 10, 2026.

If you hold your shares in *street name*, you may vote:

- **By internet***. Via the Internet at www.proxyvote.com. To vote online, you will need your unique control number;
- **By telephone***. If you are located within the United States and Canada, call 1-800-454-8683 (toll-free) from a touch-tone telephone;
- **By mail**. By returning a properly executed voting instruction form by mail, depending upon the method(s) your broker, bank or other nominee makes available; or
- **In person**. To do so, you must request a legal proxy from your broker, bank, or other nominee and present it at the Special Meeting.

**Not all street name holders may be able to vote at the web address and phone number provided above.*

If your shares are held in street name, please check the voting instruction form provided to you by your broker, bank, or other nominee for internet or telephone voting availability. If internet and/or telephone voting is available to a street name holder, such facilities will typically close at 11:59 p.m. Eastern Time on February 10, 2026; however, it may be earlier. To vote online or via telephone, you will need your unique control number. You can find the control number on your voting instruction form. Be sure to have your voting instruction form in hand and follow the instructions.

What if I return a proxy card or otherwise submit a proxy but do not make specific choices?

All shares held by recordholders entitled to vote, represented by a properly executed and unrevoked proxy received in time for the Special Meeting, will be voted in accordance with the instructions given. In the absence of such instructions, shares will be voted as recommended by the Board.

What are “broker non-votes”?

Banks, brokers, and other agents acting as nominees are permitted to use discretionary voting authority to vote for proposals that are deemed “routine” by the New York Stock Exchange, which means that they can submit a proxy or cast a ballot on behalf of stockholders who do not provide a specific voting instruction. Brokers, banks, or other nominees are not permitted to use discretionary voting authority to vote for proposals that are deemed “non-routine” by the New York Stock Exchange. Under the rules and interpretations of the New York Stock Exchange, “non-routine” matters are matters that may substantially affect the rights or privileges of stockholders, such as mergers, stockholder proposals, elections of directors (even if not contested), executive compensation, and certain corporate governance proposals, even if management supported. We believe that neither of the proposals at the Special Meeting will be treated by the New York Stock Exchange as a routine matter and both will be treated by the New York Stock Exchange as non-routine matters. The determination of which proposals are deemed “routine” versus “non-routine” may not be made by the New York Stock Exchange until after the date on which this proxy statement has been distributed. As such, it is important that you provide voting instructions to your broker, bank, or other nominee as to how to vote your shares, if you wish to ensure that your shares are present and voted at the Special Meeting on all matters.

When there is at least one “routine” matter to be considered at a meeting, a “broker non-vote” occurs when a proposal is deemed “non-routine” and a nominee holding shares for a beneficial owner does not have discretionary voting authority with respect to the “non-routine” matter being considered and has not received instructions from the beneficial owner. Since neither proposal will be deemed to be a “routine” matter broker non-votes will not occur at the Special Meeting.

What constitutes a quorum?

A quorum for the transaction of business at the Special Meeting requires representation, in person or by proxy, of the holders of one third of the Company’s voting power of the shares of stock entitled to vote at the meeting. Abstentions will be counted as shares that are present for purposes of determining the presence of a quorum for the transaction of business at the Special Meeting.

What is the voting requirement to approve each of the proposals?

Proposal	Vote Required	Voting Options	Impact of "Abstain" Vote	Impact of Broker Non-Votes
Proposal No. 1: Warrant Exercise Proposal	The affirmative vote of a majority of the shares present in person or by proxy and entitled to vote on the matter is required.	"FOR" "AGAINST" "ABSTAIN"	Counts as a vote "AGAINST" this proposal.	Not applicable-There are not expected to be any.
Proposal No. 2: Adjournment Proposal	The affirmative vote of a majority of the shares present in person or by proxy and entitled to vote on the matter is required.	"FOR" "AGAINST" "ABSTAIN"	Counts as a vote "AGAINST" this proposal.	Not applicable-There are not expected to be any.

How does the Board recommend that I vote?

The Board recommends that you vote your shares:

- **"FOR"** the Warrant Exercise Proposal (Proposal No. 1); and
- **"FOR"** the Adjournment Proposal (Proposal No. 2).

Who will count the vote?

One or more inspectors of election at the Special Meeting will tabulate and certify the votes.

What does it mean if I receive more than one set of proxy materials?

It means that your shares are held in more than one account at the transfer agent and/or with banks or brokers. Please vote all of your shares. To ensure that all of your shares are voted, please submit your proxy for each set of proxy materials via the internet, telephone or by signing, dating, and returning the enclosed proxy card and/or voting instruction card in the enclosed envelope.

May I change my vote or revoke my proxy?

Yes. Whether you have voted by internet, telephone or mail, if you are a stockholder of record, you may change your vote and revoke your proxy by:

- sending a written statement to that effect to the attention of the Company's Secretary at the Company's principal office at 9605 Medical Center Drive, Suite 270, Rockville, Maryland 20850, provided such statement is received no later than 11:59 p.m. Eastern Time on February 10, 2026;
- voting again by internet or telephone at a later time before the closing of those voting facilities at 11:59 p.m. Eastern Time on February 10, 2026;
- submitting a properly signed proxy card to the attention of the Company's Secretary at the Company's principal office at 9605 Medical Center Drive, Suite 270, Rockville, Maryland 20850, with a later date that is received no later than 11:59 p.m. Eastern Time on February 10, 2026; or
- attending and voting at the Special Meeting on February 11, 2026.

Your last vote is the one that is counted. Your attendance at the Special Meeting by itself will not revoke your proxy unless you give written notice of revocation to the Company before your proxy is voted or you vote at the Special Meeting.

If you hold shares in street name, please refer to information from your bank, broker, or other intermediary on how to revoke or submit new voting instructions.

Who will pay for the cost of this proxy solicitation?

The Company will pay for the cost of soliciting proxies. Some of the Company's directors, officers, or employees may (for no additional compensation) solicit proxies in person or by telephone, email, or facsimile transmission. Brokers and other nominees will be requested to solicit proxies or authorizations from beneficial owners and will be reimbursed for their reasonable expenses. In addition, we have retained D.F. King & Co., Inc. to aid in the solicitation of proxies for the Special Meeting. We will pay D.F. King & Co., Inc. fees of not more than \$8,000 plus expense reimbursement for its services. Please contact (800) 859-8509 with any questions you may have regarding our proposals. In addition to the use of the mail, proxies may be solicited by personal interview, telephone, telegram, facsimile and advertisement in periodicals and postings, in each case by our directors, officers and employees without additional compensation.

PROPOSAL NO. 1
APPROVAL OF THE WARRANT EXERCISE

We are seeking stockholder approval for the issuance of up to 16,184,560 shares of our Common Stock upon the exercise of common stock purchase warrants that were issued in connection with our private placement offering (the “**Private Placement**”) that closed on October 17, 2025, as contemplated by Section 713(a) of the NYSE American Company Guide and the terms of an inducement agreement.

On October 16, 2025, we entered into a warrant inducement agreement (the “**Inducement Agreement**”) with certain institutional investors (the “**Holders**”) of outstanding warrants to purchase shares of Common Stock issued in a private placement offering that closed on September 27, 2024 (the “**September Warrants**”) and a private placement offering that closed on May 8, 2025 (the “**May Warrants**”) and, together with the September Warrants, collectively, the “**Existing Warrants**”). Pursuant to the Inducement Agreement, the Holders of the Existing Warrants exercised for cash, at a reduced exercise price of \$0.54 per share, September Warrants to purchase an aggregate of up to 1,345,000 shares of Common Stock (the “**September Warrant Shares**”) and May Warrants to purchase an aggregate of up to 6,747,280 shares of Common Stock (the “**May Warrant Shares**”) and, together with the September Warrant Shares, the “**Existing Warrant Shares**”) and, in consideration of the Holders’ immediate exercise of the Existing Warrants in accordance with the Inducement Agreement, we issued unregistered Common Stock Purchase Warrants (the “**New Warrants**”) to purchase an aggregate of up to 16,184,560 shares of Common Stock (200% of the number of Existing Warrant Shares issued upon exercise of the Existing Warrants) (the “**New Warrant Shares**”), at an exercise price of \$0.54, to the Holders of the Existing Warrants. The transactions contemplated by the Inducement Agreement (the “**Warrant Inducement**”) were consummated on October 17, 2025 (the “**Closing Date**”).

The New Warrants are exercisable commencing on the effective date of stockholder approval (the “**Stockholder Approval**”) for the full exercise of the Warrant and issuance of all of the shares of Common Stock issuable upon exercise of the New Warrants (the “**Stockholder Approval Date**”) and expire on the fifth anniversary of the Stockholder Approval Date. If at any time there is no effective registration statement under the Securities Act for the resale of the New Warrant Shares, any holder may, in its sole discretion, elect to exercise New Warrants through a cashless exercise, in which case such holder would receive upon such exercise the net number of shares of Common Stock determined according to the formula set forth in the New Warrant. We also agreed to use our reasonable best efforts to obtain all components of the Stockholder Approval. If we do not obtain all components of the Stockholder Approval at the first meeting of our stockholders called within sixty (60) days following the Closing Date, we are obligated to call a meeting every sixty (60) days thereafter to seek any remaining components of the Stockholder Approval until the earlier of the date on which Stockholder Approval is obtained or the New Warrants are no longer outstanding. A special meeting had originally been scheduled for December 15, 2025 for the purpose of seeking Stockholder Approval. On December 15, 2025, we called to order such special meeting. At that time, there were not present or represented by proxy a sufficient number of shares of the Company’s common stock to constitute a quorum. As a result, we were not able to convene such special meeting and obtain Stockholder Approval at such time.

The exercise price of the New Warrants, and the number of New Warrant Shares, are subject to adjustment in the event of any stock dividend or split, reverse stock split, recapitalization, reorganization, or similar transaction, as described in the New Warrants. In addition, the New Warrants provide that we may also, at any time during the term of the New Warrants, subject to the prior written consent of the applicable holder, voluntarily reduce the then current exercise price to any amount and for any period of time, subject to the rules and regulations of NYSE American. Accordingly, approval of this proposal is also approval of the issuance of the New Warrant Shares at a lower exercise price than the initial exercise price.

A holder will not have the right to exercise any portion of the New Warrants if the holder (together with its affiliates) would beneficially own in excess of 4.99% (or, upon election of the holder, 9.99%) of the number of shares of our Common Stock outstanding immediately after giving effect to the exercise, as such percentage ownership is determined in accordance with the terms of the New Warrants. However, any holder may increase or decrease such percentage, provided that any increase will not be effective until the 61st day after such election.

In the event of a Fundamental Transaction (as such term is defined in the New Warrant), then the successor entity will succeed to, and be substituted for us, and may exercise every right and power that we may exercise and will assume all of our obligations under the New Warrants with the same effect as if such successor entity had been named in the warrant itself. If holders of Common Stock are given a choice as to the securities, cash, or property to be received in a Fundamental Transaction, then the holder shall be given the same choice as to the consideration it receives upon any exercise of the New Warrants following such Fundamental Transaction. In addition, the successor entity, at the request of holders of New Warrants, will be obligated to purchase any unexercised portion of the New Warrants in accordance with the terms thereof. Notwithstanding the foregoing, in the event of a Fundamental Transaction, the holders of the New Warrants have the right to require us or a successor entity to redeem the New Warrants for cash in the amount of the Black Scholes Value (as defined in the New Warrant) of the unexercised portion of the New Warrants concurrently with or within 30 days following the consummation of a Fundamental Transaction. However, in the event of a Fundamental Transaction which is not in our control, including a Fundamental Transaction not approved by our Board, the holders of the New Warrants will only be entitled to receive from us or our successor entity, as of the date of consummation of such Fundamental Transaction, the same type or form of consideration (and in the same proportion), at the Black-Scholes Value of the unexercised portion of the New Warrant that is being offered and paid to the holders of Common Stock in connection with the Fundamental Transaction, whether that consideration is in the form of cash, stock or any combination of cash, and stock, or whether the holders of our Common Stock are given the choice to receive alternative forms of consideration in connection with the Fundamental Transaction.

We agreed in the Inducement Agreement to file a registration statement to register the resale of the New Warrant Shares (the “**Resale Registration Statement**”) as soon as practicable (and in any event within 30 calendar days following the date of the Inducement Agreement), and to use commercially reasonable efforts to have such Resale Registration Statement declared effective by the Securities and Exchange Commission (the “**Commission**”) within 60 days following the date of the Inducement Agreement (or 90 days following the date of the Inducement Agreement in the event of a “limited review” or “full review” by the Commission), provided, however, that if the Commission is closed for operations due to a government shutdown, the applicable deadline for the Resale Registration Statement becoming effective shall be extended by the same amount of days that the Commission remains closed for operations, and to keep the Resale Registration Statement effective at all times until no holder of the New Warrants owns any New Warrants or New Warrant Shares. We filed a registration statement on Form S-3 registering the Warrant Shares for resale on November 7, 2025, which registration statement was declared effective on December 2, 2025.

Pursuant to the Inducement Agreement, we also agreed, with certain limited exceptions, for 45 days following the Closing Date, not to issue any shares of Common Stock or Common Stock Equivalents (as defined in the New Warrants) or to file any registration statement or prospectus, subject to certain exceptions. We also agreed not to enter into a Variable Rate Transaction (as defined in the Inducement Agreement) for a period of 90 days following the Closing Date, provided however that sales pursuant to an at-the-market are permitted to be made beginning five (5) days after Closing Date.

A.G.P./Alliance Global Partners LP (“**AGP**”) served as our exclusive financial advisor in connection with the Warrant Exercise and other transactions described in the Inducement Agreement. Pursuant to the terms of an engagement letter, dated October 16, 2025, by and between us and AGP (the “**Engagement Letter**”), we paid to AGP a cash fee equal to 7.0% of the aggregate gross proceeds received from the Holders upon exercise of the Existing Warrants and reimbursed certain expenses.

Reasons for the Warrant Exercise Proposal

Our Common Stock is listed on NYSE American and trades under the ticker symbol “TOVX.” Section 713(a) of the NYSE American Company Guide requires stockholder approval of transactions other than public offerings of greater than 20% of the outstanding common stock or voting power of an issuer prior to a private placement for less than the applicable Minimum Price. Under NYSE American Company Guide, the “**Minimum Price**” means a price that is the lower of: (i) the closing price immediately preceding the signing of the binding agreement; or (ii) the average closing price of the Common Stock for the five trading days immediately preceding the signing of the binding agreement. The closing price of our Common Stock on NYSE American on October 15, 2025, the trading date immediately preceding the signing of the Inducement Agreement, was \$0.84 per share and the average closing price of our Common Stock for the five trading days immediately preceding the signing of the Inducement Agreement was \$0.44. In order to comply with Section 713(a) of the NYSE American Company Guide, and in compliance with the terms of the Inducement Agreement, the New Warrants are not exercisable until Stockholder Approval is obtained.

We are seeking stockholder approval for the issuance of up to an aggregate of 16,184,560 shares of our Common Stock upon the exercise of the New Warrants. Effectively, stockholder approval of this Warrant Exercise Proposal is one of the conditions for us to receive up to approximately \$8.7 million in gross proceeds upon the exercise of the 16,184,560 New Warrants, if exercised for cash. Loss of these potential funds could adversely impact our ability to fund our operations.

The Board is not seeking the approval of our stockholders to authorize our entry into or consummation of the Warrant Inducement, as the Warrant Inducement has already been completed. We are only asking for approval to issue up to an aggregate of 16,184,560 New Warrant Shares upon exercise of the New Warrants.

Potential Consequences if Proposal No. 1 is Not Approved

The failure of our stockholders to approve this Proposal No. 1 will mean that: (i) we cannot permit the exercise of the New Warrants and (ii) may incur substantial additional costs and expenses because we are required to hold additional stockholder meetings until the earlier of the date on which Stockholder Approval is obtained or the New Warrants are no longer outstanding.

Each New Warrant has an initial exercise price of \$0.54 per share. Accordingly, if this proposal is not approved we would not be able to realize the gross proceeds from the exercise of the New Warrants. If all the New Warrants were exercised for cash we would realize an aggregate of up to approximately \$8.7 million in gross proceeds. However, if we were to lower the exercise price of the New Warrants we will receive less proceeds.

Potential Adverse Effects of the Approval of Proposal No. 1

If this Proposal No. 1 is approved, existing stockholders will suffer dilution in their ownership interests in the future upon the issuance of the New Warrant Shares upon exercise of the New Warrants. Assuming the full exercise of the New Warrants, an aggregate of 16,184,560 additional shares of Common Stock will be outstanding, and the ownership interest of our existing stockholders would be correspondingly reduced. In addition, the sale into the public market of these shares also could materially and adversely affect the market price of our Common Stock.

Required Vote

The affirmative vote from the holders of a majority of the shares present in person or represented by proxy and entitled to vote on the Warrant Exercise Proposal at the Special Meeting is required for approval of this proposal. Abstentions will have the same effect as votes AGAINST this proposal. Broker non-votes are not expected for this proposal.

**THE BOARD UNANIMOUSLY RECOMMENDS THAT
YOU VOTE "FOR" THE WARRANT EXERCISE PROPOSAL**

**PROPOSAL NO. 2
THE ADJOURNMENT PROPOSAL**

Background of and Rationale for the Adjournment Proposal

The Board of Directors believes that if at the Special Meeting, the number of votes represented by shares of our Common Stock, present or represented and voting in favor of the Warrant Exercise Proposal, is insufficient to approve the Warrant Exercise Proposal or establish a quorum, it is in the best interests of the stockholders to enable the Board to continue to seek to obtain a sufficient number of additional votes to approve the Warrant Exercise Proposal.

In the Adjournment Proposal, we are asking stockholders to authorize the holder of any proxy solicited by the Board to vote in favor of adjourning or postponing the Special Meeting or any adjournment or postponement thereof. If our stockholders approve this proposal, we could adjourn or postpone the Special Meeting, and any adjourned session of the Special Meeting, to use the additional time to solicit additional proxies in favor of the Warrant Exercise Proposal.

Additionally, approval of the Adjournment Proposal could mean that, in the event we receive proxies indicating that holders of a majority of the number of shares present in person or represented by proxy at the Special Meeting will vote against the Warrant Exercise Proposal, we could adjourn or postpone the Special Meeting without a vote on the Warrant Exercise Proposal and use the additional time to solicit the holders of those shares to change their vote in favor of the Warrant Exercise Proposal.

Vote Required

The affirmative vote from the holders of a majority of the shares present in person or represented by proxy and entitled to vote on the Adjournment Proposal at the Special Meeting is required for approval of this proposal. Abstentions will have the same effect as a vote AGAINST this proposal. Broker non-votes are not expected for this proposal.

**THE BOARD OF DIRECTORS UNANIMOUSLY RECOMMENDS A
VOTE "FOR" THE ADJOURNMENT PROPOSAL.**

SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT

The following table sets forth information, as of the Record Date, or as otherwise set forth below, with respect to the beneficial ownership of our Common Stock (i) all persons known to us to be the beneficial owners of more than 5% of the outstanding shares of our Common Stock; (ii) each of our directors and our named executive officers named in the Summary Compensation Table; and (iii) all of our directors and our current executive officer as a group.

Name and Address of Beneficial Ownership(1)	Shares Owned(2)	
	Number of Shares Owned	Percentages of Shares(3)
Named Executive Officers and Directors		
Jeffrey J. Kraws(4)	31,659	*
Steven Shallcross(5)	136,908	*
Jeffrey Wolf(6)	31,633	*
John Monahan(7)	30,733	*
All current officers and directors as a group (4 persons)	230,933	

* represents less than 1% of our Common Stock

(1) The address for each officer and director is 9605 Medical Center, Suite 270, Rockville, Maryland 20850.

(2) Beneficial ownership is determined in accordance with SEC rules and generally includes voting or investment power with respect to securities. Except as indicated in the footnotes to the table, to the knowledge of the Company, the persons named in the table have sole voting and investment power with respect to all shares of Common Stock, preferred stock, options and/or warrants shown as beneficially owned by them, subject to community property laws, where applicable. Pursuant to the rules of the Commission, the number of shares of our Common Stock deemed outstanding includes shares issuable pursuant to options held by the respective person or group that are currently exercisable or may be exercised within 60 days of the Record Date.

(3) As of the Record Date, the Company had 35,688,349 shares of Common Stock outstanding.

(4) Includes 31,633 shares issuable upon exercise of options held by Mr. Kraws that are exercisable within the 60-day period following the Record Date. Does not include an additional 4,167 shares issuable upon exercise of options held by Mr. Kraws issuable upon exercise of options held by Mr. Kraws that are not exercisable within the 60-day period following the Record Date.

(5) Includes 126,908 shares issuable upon exercise of options held by Mr. Shallcross and 28,704 shares of Common Stock issuable upon exercise of options held by Mrs. Shallcross (Mr. Shallcross's wife) that are exercisable within the 60-day period following the Record Date. Does not include an additional 145,001 shares issuable upon exercise of options held by Mr. Shallcross that are not exercisable within the 60-day period following the Record Date.

(6) Includes 31,633 shares issuable upon exercise of options held by Mr. Wolf that are exercisable within the 60-day period following the Record Date. Does not include an additional 4,167 shares issuable upon exercise of options held by Mr. Wolf issuable upon exercise of options held by Mr. Wolf that are not exercisable within the 60-day period following the Record Date.

(7) Includes 30,733 shares issuable upon exercise of options held by Dr. Monahan that are exercisable within the 60-day period following the Record Date. Does not include an additional 4,167 shares issuable upon exercise of options held by Dr. Monahan issuable upon exercise of options held by Dr. Monahan that are not exercisable within the 60-day period following the Record Date.

STOCKHOLDER PROPOSALS FOR THE 2026 ANNUAL MEETING

Stockholders who intend to present proposals for inclusion in the proxy materials for the 2026 Annual Meeting of Stockholders (the “2026 Annual Meeting”) under SEC Rule 14a-8 must ensure that such proposals are received by the Corporate Secretary of the Company not later than March 11, 2026. Such proposals must meet the requirements of the Commission to be eligible for inclusion in our 2026 proxy materials.

If you wish to nominate a director at the 2026 Annual Meeting, or to bring any other proposal before the 2026 Annual Meeting, that is not to be included in the 2026 Annual Meeting proxy materials pursuant to Rule 14a-8, you must do so in accordance with our Bylaws, which requires notice of not less than 120 days prior to the anniversary of the preceding year’s annual meeting to be timely, which in this case means May 1, 2026, provided, however, that in the event that the date of the annual meeting is changed by more than 30 days from such anniversary date, notice by the stockholder to be timely must be so received not later than the close of business on the tenth day following the earlier of the day on which notice of the date of the meeting was mailed or public disclosure was made. The stockholder’s notice must set forth the information required by our Bylaws with respect to each stockholder making the proposal and each proposal and director nomination that such stockholder intends to present at the 2026 Annual Meeting. All stockholder director nominations must be given in writing to the Chairman of the Nominations Committee, Theriva Biologics, Inc., 9605 Medical Center Drive, Suite 270, Rockville, Maryland 20850, All other stockholder proposals must be given in writing to the Corporate Secretary, Theriva Biologics, Inc., 9605 Medical Center Drive, Suite 270, Rockville, Maryland 20850.

In addition to satisfying the foregoing requirements, to comply with the universal proxy rules, stockholders who intend to solicit proxies in support of director nominees other than the Company’s nominees must provide notice that sets forth the information required by Rule 14a-19 under the Exchange Act no later than June 30, 2026. If such meeting date is more than 30 days before or after August 29, 2026, then notice must be provided by the later of 60 calendar days prior to the date of the annual meeting or the 10th calendar day following the day on which public announcement of the date of the annual meeting is first made.

HOUSEHOLDING

The Commission has adopted rules that permit companies and intermediaries (e.g., brokers) to satisfy the delivery requirements for proxy statements, annual reports, and notices of internet availability of proxy materials with respect to two or more stockholders sharing the same address by delivering a single copy of the applicable document(s) addressed to those stockholders. This process, which is commonly referred to as “householding,” potentially means extra convenience for stockholders and cost savings for companies.

Brokers with account holders who are stockholders of the Company may be “householding” our proxy materials. A single proxy statement or notice may be delivered to multiple stockholders sharing an address unless contrary instructions have been received from the affected stockholders. Once you have received notice from your broker that it will be “householding” communications to your address, “householding” will continue until you are notified otherwise or until you notify your broker or the Company that you no longer wish to participate in “householding.”

Street-name stockholder who are currently receiving househanded materials may revoke their consent, and street-name shareholders who are not currently receiving househanded materials may request householding of our future materials, by contacting Broadridge Financial Services, Inc., either by calling toll free at (866) 540-7095 or by writing to Broadridge Household Department, 51 Mercedes Way, Edgewood, New York 11717. If you revoke your consent you will be removed from the “householding” program within 30 days of Broadridge’s receipt of your revocation, and each stockholder at your address will receive individual copies of our future materials.

By Order of the Board of Directors:

/s/ Steven A. Shallcross

Steven A. Shallcross

Chief Executive Officer, Chief Financial Officer and Director



THERIVA BIOLOGICS, INC.
9605 MEDICAL CENTER DRIVE, SUITE 270
ROCKVILLE, MARYLAND 20850



SCAN TO
VIEW MATERIALS & VOTE



VOTE BY INTERNET - www.proxyvote.com or scan the QR Barcode above
Use the Internet to transmit your voting instructions and for electronic delivery of information up until 11:59 p.m. Eastern Time the day before the meeting date. Have your proxy card in hand when you access the web site and follow the instructions to obtain your records and to create an electronic voting instruction form.

ELECTRONIC DELIVERY OF FUTURE PROXY MATERIALS
If you would like to reduce the costs incurred by our company in mailing proxy materials, you can consent to receiving all future proxy statements, proxy cards and annual reports electronically via e-mail or the Internet. To sign up for electronic delivery, please follow the instructions above to vote using the Internet and, when prompted, indicate that you agree to receive or access proxy materials electronically in future years.

VOTE BY PHONE - 1-800-690-6903
Use any touch-tone telephone to transmit your voting instructions up until 11:59 p.m. Eastern Time the day before the meeting date. Have your proxy card in hand when you call and then follow the instructions.

VOTE BY MAIL
Mark, sign and date your proxy card and return it in the postage-paid envelope we have provided or return it to Vote Processing, c/o Broadridge, 51 Mercedes Way, Edgewood, NY 11717.

TO VOTE, MARK BLOCKS BELOW IN BLUE OR BLACK INK AS FOLLOWS:

V82661-530300

KEEP THIS PORTION FOR YOUR RECORDS
DETACH AND RETURN THIS PORTION ONLY

THIS PROXY CARD IS VALID ONLY WHEN SIGNED AND DATED.

THERIVA BIOLOGICS, INC.

The Board of Directors recommends you vote **FOR** proposals 1 and 2:

- | | For | Against | Abstain |
|--|--------------------------|--------------------------|--------------------------|
| 1. To approve, pursuant to Section 713(a) of the NYSE American Company Guide and the terms of an inducement agreement, the issuance of up to an aggregate of 16,184,560 shares of our common stock upon the exercise of our common stock purchase warrants issued to institutional investors in our private placement offering, which closed on October 17, 2025 (the " Warrant Exercise Proposal "). | ☐ | ☐ | ☐ |
| 2. To approve a proposal to adjourn the Special Meeting to a later date, if necessary or appropriate, to permit further solicitation and vote of proxies in the event that there are insufficient votes for, or otherwise in connection with, the approval of the Warrant Exercise Proposal (the " Adjournment Proposal "). | ☐ | ☐ | ☐ |

Note: To transact such other business as may properly come before the meeting or any adjournments or postponements of the meeting.

Please sign exactly as your name(s) appear(s) hereon. When signing as attorney, executor, administrator, or other fiduciary, please give full title as such. Joint owners should each sign personally. All holders must sign. If a corporation or partnership, please sign in full corporate or partnership name by authorized officer.

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Signature [PLEASE SIGN WITHIN BOX]

Date

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Signature (Joint Owners)

Date

Important Notice Regarding the Availability of Proxy Materials for the Special Meeting:
The Notice of Meeting, Proxy Statement and Proxy Card are available at www.proxyvote.com.

V82662-S30300

**THERIVA BIOLOGICS, INC.
SPECIAL MEETING OF STOCKHOLDERS
FEBRUARY 11, 2026, AT 3:30 P.M. (LOCAL TIME)
THIS PROXY IS SOLICITED ON BEHALF OF THE BOARD OF DIRECTORS**

The undersigned stockholder hereby appoints Steven A. Shallcross as proxy, with the power to appoint his substitute, and hereby authorizes him to represent and to vote, as designated on the reverse side of this ballot, all of the shares of common stock of THERIVA BIOLOGICS, INC. that the undersigned is entitled to vote at the Special Meeting of Stockholders to be held at 3:30 p.m. (local time) on February 11, 2026, at the Company's principal place of business for research and clinical development, located at Carrer Torrent de Can Ninou, naus 5-6, 08150 - Parets del Vallès (Barcelona) Spain, or any adjournment or postponement thereof.

This proxy, when properly executed, will be voted in the manner directed herein. If no such direction is made, this proxy will be voted in accordance with the Board of Directors' recommendations.

Continued and to be signed on reverse side